

AHMAD ARRABI

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RESEARCH OVERVIEW

I am a 4th-year Ph.D. candidate in Computer Science at the University of Vermont. My research focuses on automating intraoperative X-ray systems like biplanes and C-arms. I bridge the gap between computer vision research and clinical practice. By combining AI expertise with an understanding of interventional workflows, my research prioritizes measurable clinical solutions with the end goal of operating room deployment.

EDUCATION

Ph.D. in Computer Science <i>University of Vermont</i>	2023 – 2027 (expected) <i>Advisor: Safwan Wshah</i>
M.Sc. in Computer Science <i>University of Vermont</i>	2023 – 2025
B.S. in Computer Engineering <i>Princess Sumaya University for Technology</i>	2017 – 2022

INDUSTRY EXPERIENCE

Research Collaboration Intern <i>Siemens Healthineers in Collaboration with the Cleveland Clinic Foundation</i>	Summer 2026
- Managed the end-to-end research pipeline from clinical problem formulation and data curation to simulation development and deep learning evaluation on historical clinical data - Worked with hospital clinical data systems including DICOM imaging, Epic electronic health records, and REDCap databases within HIPAA-compliant environments - Engineered a differentiable DRR simulation using diffDRR for large-scale synthetic X-ray generation (~6 million training DRRS) and 6-DoF C-arm pose estimation - Conducted a retrospective analysis on ~800 neurointerventional procedures to quantify radiation exposure and procedure time, establishing requirements for automated biplane positioning - Demonstrated that AI-driven automated positioning can address long-tail clinical edge cases, projecting over 90% reduction in radiation exposure	

RESEARCH EXPERIENCE

Graduate Research Assistant <i>University of Vermont – Vermont Artificial Intelligence Laboratory (VaiL)</i>	2023 – Present
- Routinely developed multi-GPU training pipelines on HPC clusters using Slurm across NVIDIA CUDA and AMD ROCm environments through the AMD University Program - Drove research projects from problem formulation through dataset development, model design, large-scale experimentation, evaluation, publication, and open-source release	
Medical Imaging: - Collaborated with interventional radiologists at the Cleveland Clinic and UVMHC to translate procedural C-arm positioning requirements into deep learning-based solutions for autonomous localization - Led the development of a self-supervised framework for autonomous C-arm localization from intraoperative X-rays - Designed an uncertainty-aware control pipeline for C-arm servoing toward predefined anatomical targets - Fine-tuned multimodal large language models for skeletal landmark localization and integrated them into an agentic C-arm control pipeline for spatial reasoning and target-driven navigation	
Generative and Multimodal Learning: - Designed a diffusion bridge framework (with Diffusers and Hugging Face) for joint multimodal generation through controllable transport between diffusion trajectories during sampling - Adapted large-scale diffusion and vision-language foundation models for task-specific objectives, including interleaved co-generation, controllable image editing, and multimodal synthesis - Developed geometry-guided conditioning strategies (with ControlNet) for cross-view diffusion under extreme view-point shifts, reducing domain gap in ground-to-aerial generation	

Conferences

- **Arrabi, A.**, Zhang, X., Li, X., & Wshah, S., (2026). Split and Bridge: Multimodal Generation via Diffusion Bridging. In Advances in Neural Information Processing Systems (NeurIPS) ([Accepted](#))
- **Arrabi, A.**, Jung, J., Le, J., Nguyen, A., Reed, J., Stahl, E., Franssen, N., Raymond, S., & Wshah, S. (2025). C-ARM Guidance: A Self-Supervised Approach to Automated Positioning During Stroke Thrombectomy. In 2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI) (pp. 1-4)
- **Arrabi, A.**, Zhang, X., Sultani, W., Chen, C., & Wshah, S. (2025). Cross-View Meets Diffusion: Aerial Image Synthesis with Geometry and Text Guidance. In 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) (pp. 5356-5366)
- Chen, Y., Zhang, X., **Arrabi, A.**, Sultani, W., Chen, C., & Wshah, S. (2026). Turning Generators into Retrievers: Unlocking MLLMs for Natural Language-Guided Geo-Localization. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings
- Lehyeh, A. A., Zhang, X., **Arrabi, A.**, Sultani, W., Chen, C., & Wshah, S. (2026). GeoFlow: Real-Time Fine-Grained Cross-View Geolocalization via Iterative Flow Prediction. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

Peer-Reviewed Workshops

- **Arrabi, A.**, Jung, J., Luo, J., Franssen, N., Raymond, S., & Wshah, S. (2025). Automated C-Arm Positioning via Conformal Landmark Localization. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops (pp. 4392-4401)

Journals

- Jung, J.H., **Arrabi, A.**, Luo, J., Raymond, S.B., & Wshah, S. Autonomous skeletal landmark localization toward agentic C-arm control. Int J CARS 21, 1015–1023 (2026), ([presented at IPCAI 2026](#))
- Al-Mousa, A., **Arrabi, A.**, & Daoud, H. (2025). A reinforcement learning-based reverse-parking system for autonomous vehicles. IET Intelligent Transport Systems, 19(1), e12614

Abstracts

- **Arrabi, A.**, Luo, J., Von Roden, M., Jung, J.H., Wshah, S. & Raymond, S.B., (2026). Vision-Language Models Trained on Synthetic DRRs Generalize to Clinical Fluoroscopy. Unlocking Artificial Intelligence, AI Radiology Symposium, Hollywood, FL.
- Kashkoush A., **Arrabi A.**, Jung J., Franssen N., Wshah S., Raymond S. (2025). Deep learning model predicts biplane trajectory from pelvis to head. Society for NeuroInterventional Surgery, 7/2025.

ACADEMIC SERVICES & AWARDS

Reviewing:

- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026.
- International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), 2026.
- International Journal of Computer Assisted Radiology and Surgery (IJCARS)
- Pattern Recognition
- Pattern Recognition Letters
- European Conference on Artificial Intelligence (ECAI), 2025.

Invited Talks:

- International Workshop on Video Surveillance Systems in Smart Cities, WACV 2025.

Awards:

- MICCAI Society Membership Grant, 2026.